DEP SPECIFICATION

STRUCTURAL DESIGN AND ENGINEERING OF ONSHORE STRUCTURES

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DESIGN AND ENGINEERING PRACTICE

PREFACE

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APPENDICES

1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the design of onshore structural assets (structures).

In this DEP, safety is established as the primary consideration for structural design.

This is a revision of the DEP of the same number dated February 2015; see (1.5) regarding the changes.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this DEP is confined to Shell companies and, where necessary, to Contractors and Manufacturers/Suppliers nominated by them. Any authorized access to DEPs does not for that reason constitute an authorization to any documents, data or information to which the DEPs may refer.

This DEP is intended for use in facilities related to oil and gas production, gas handling oil refining, chemical processing, gasification, distribution and supply/marketing. This DEP may also be applied in other similar facilities.

When DEPs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this DEP, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned, the objective being to obtain agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.4 CROSS-REFERENCES

Where cross-references to other parts of this DEP are made, the referenced section or clause number is shown in brackets ( ). Other documents referenced by this DEP are listed in (6).

1.5 SUMMARY OF MAIN CHANGES

This DEP is a revision of the DEP of the same number dated February 2015. The changes followed a requirements quality review and were not a full technical update. The following are the main, non-editorial changes.

Section/Clause	**Change**
All	Restructured and clarified requirements, enforced use of shall/should/may for normative statements.

1.6 COMMENTS ON THIS DEP

Comments on this DEP may be submitted to the Administrator using one of the following options:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a DEP and then go to the details screen for that DEP. Click on the "Give feedback" link, fill in the online form and submit.
DEP Feedback System (Users with access to Shell Wide Web)	Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal http://sww.shell.com/standards. Select "Submit DEP Feedback", fill in the online form and submit.
DEP Standard Form (other users)	Use DEP Standard Form 00.00.05.80-Gen. to record feedback and email the form to the Administrator at standards@shell.com.

Feedback that has been registered in the DEP Feedback System by using one of the above options will be reviewed by the DEP Custodian for potential improvements to the DEP.

1.7 DUAL UNITS

This DEP contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this DEP indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant DEP clause.

1.9 PROJECT DOCUMENTATION REQUIREMENTS

For projects to ensure compliance with this DEP, proper documentation is required in the Definition and Execution phases of the project.

2. DESIGN PROCESS AND DELIVERABLES

2.1 APPLICABLE DESIGN SPECIFICATIONS

1. The Code, standard or practice for design load as specified in Table 3.1 shall be used in combination with the corresponding code, standard or practice for design criteria specified in Table 5.1.
   a. The latest edition of the applicable code, standard or practice that is in effect on the date of contract award shall be used, except as otherwise noted.

   Full titles of the applicable codes, standards and practices are listed in (6).

2.2 DESIGN STAGE AND REQUIREMENTS

1. Refer to (Appendix A) for requirements of engineering design documentation during various phases of structural design.
2. Refer to (Appendix B) for requirements of Structural engineering drawings.

2.3 MODULE STRUCTURAL DESIGN

2.3.1 General

1. The module and materials shall be designed for the proposed loadin and loadout, transportation and installation conditions in accordance with applicable codes or standards in Table 3.1 and Table 5.1, in line with project requirements.
   a. Module structural members shall be strengthened where necessary.
2. A weight management plan and report shall be developed and updated throughout the project, and include the following:
   a. module weight calculations including the following;
      i. centre of gravity calculations for each step in transportation process;
      ii. centre of gravity for lifting ,erection and installation (i.e., with all temporary transportation steel removed prior to lifting, and any additional ship loose item installed prior to lifting).
   b. load out and Transportation support points;
   c. lift points for lifting, erection and installation;
   d. overall Module dimensions.
3. Sea fixings and localised structural reinforcement shall be provided where required to accommodate the intended transportation loads and designed to achieve stable transportation.
4. Lifting supporting points shall be designed to conform to the planned Module placement and Lifting schemes.
5. To perform the lifting analysis of the modules, the Contractor shall use approved weight and COG envelope based on weight control reports or as follows:
   a. in the absence of COG envelope, a COG inaccuracy factor of 1.1 shall be applied on the lifted weight.
6. Where information is not available to determine the final lifting weight, a minimum contingency factor of 5% shall be applied to the lifted weight.
7. Effective moments on structure shall be considered due to possible shift of centre of gravity.
8. For lifting design, a nominal transverse load of 5% of the sling force shall be applied at the centre of the pinhole simultaneously with the sling force.

9. Lifting/jacking lugs/padeye and all structural members transmitting lifting forces within the structure and directly connected to the lifting/jacking lugs/padeye shall be designed for a minimum impact factor of 2.0.
10. All other structural members transmitting lifting forces shall be designed using impact factor of 1.3.
11. The design load for shackles and other rigging gear shall be based on a minimum impact factor of 1.3 on the working load.
12. All material used for fabrication of lifting lugs and padeyes shall be procured directly from steel mills approved by Principal with certification as follows:
    a. original mill certificates supplied with the fittings;
    b. original mill certificates included as part of project documentation and handed over to the Principal.
13. Full traceability of material used for lifting lugs and padeyes shall be ensured during the entire fabrication process.
14. Pad eye plates greater than 38 mm (1.5 in) thick shall be subject to through-thickness testing to ensure laminations do not exist.
    a. This requirement for through-thickness testing shall be explicitly indicated in the drawings.
15. 100% non-destructive testing (NDT) shall be performed for all the welds of the padeye as follows:
    a. ultrasonic testing (UT) for complete joint penetration welds;
    b. either dye penetrant testing or magnetic particle testing (PT/MT) for fillet welds.
16. Requirements of material properties and weld inspection, shall be explicitly shown on structural design drawings.
17. Any damage to coating of permanent steel during loadout, transportation and lifting shall be locally rectified in accordance with an approved repair procedure, after all installation is completed.
18. Any subsequent lifting of the module requiring reuse of lifting lugs and padeyes shall be shall be subject to Principal’s approval.
    a. In case of reuse of lifting lugs and padeyes, all the welds of the lifting lugs and padeyes (including connections to the main structural steel) shall undergo close visual inspection and subsequent 100% NDT as (2.3.1, Item 15).

2.3.2 Module Layout

1. The layout, size, column spacing and weight of the module should be determined accounting for all of the following:
    a. personnel access and egress requirements;
    b. human factors engineering requirements;
    c. material handling studies including provision of suitable lifting and handling device;
    d. elevated floor areas designed for maintenance and loading;
    e. grating thickness requirements to allow for heavier load movement for maintenance equipment;
    f. clearances for equipment removal and replacement;
    g. identify the fall zones of tools/equipment being lifted and the need for dropped object protection;
    h. alignment of the module structural support columns spacing and location of support points with the intended foundation philosophy (e.g., pile foundation, footing);
    i. height of module with respect to ground level based on bottom clearances, footing height, installation philosophy;
    j. logistic constraints such as height, length, breadth and weight limitations existing along the proposed route of module transportation from the fabrication yard to the final location of the module at the project site.

2.3.3 Transportation

1. For sea transportation, studies should be performed to determine the most onerous acceleration loadings, based on the intended location(s) of the module on the barge (e.g., centre, fore and aft port, fore and aft centre, fore and aft starboard).
2. The centre of gravity (COG) of each module should be kept low to enable stable transport of modules from yard to the installed location.
3. Seafastener design should be optimized to reduce shipping weights, vessel/barge dimensions and removal works at site.
4. The structural steel (e.g., transportation bracing) should be designed for easy removal after module installation.
5. Before leaving the yard, the module should be weighed in the yard using calibrated jacks to confirm the module weight and COG calculations.
    a. Weighing procedures should be submitted in advance for review and actual weighing should witnessed by the Principal or its nominated agency.

2.3.4 Lifting operations

1. Pad eye connection details should avoid tension being applied through the thickness of a plate or other rolled element.

   Lamination faults (lamellar tearing) resulted from weld shrinkage could cause a failure.

2. Pad eye main plates should be slotted through the horizontal plates and welded flat against webs or stiffening plates below.
3. Pad eye plate should be aligned with the direction of the sling as per the proposed lifting plan.
4. There should be no more than one cheek plate on each side of the main plate:
   a. the cheek plates should be circular with a radius equal to that of the main plate less the cheek plate thickness;
   b. the cheek plate thickness should be less than or equal to the main plate thickness.
5. The pinhole should meet the following:
   a. not be more than 4 mm (1/8 in) larger than the shackle pin;
   b. machined, not flame cut;
   c. line-bored after welding the cheek plates to the main plate.
6. The total thickness of the padeye should be 6 mm (1/4 in) smaller than the minimum width of the shackle.
7. Shackles should have a nominal factor of safety of 5:1 being the Minimum Breaking Load (MBL) : Working Load Limit (WLL).
8. Padeyes should not be removed from the modules unless they are interfering with the rest of the functionality of the module.

3. STRUCTURE DESIGN LOADS

3.1 GENERAL

1. The loading design basis shall be as specified in the applicable codes, standards, specifications, practices referred in Table 3.1 below unless otherwise specified by the Principal.

Table 3.1 Applicable Codes, Standards and Practices for Structure Design Loads[1)]

Design Loads	European Codes and Standards	American Codes and Standards[2)]	Canadian Codes and Standards[2)]
General Loads and Load Combination	EN 1991	ASCE/SEI 7-05 IBC) ASCE Wind Loads and Anchor Bolt Design for Petrochemical Facilities	NBC CAN/CSA-S16-01
Crane and Hoist Loads	EN 1993-6[3)] BS 6349-2 (for jetties)	OSHA 1910.179[3)] AASHTO CMAA No. 70 CMAA No. 74	AASHTO[3)] CMAA No. 70 CMAA No. 74
Erection Loads	EN 1991-1.6	ASCE/SEI 37-02	ASCE/SEI 37-02
Impact Loads	EN 1991-1.7	AASHTO[3)]	
Seismic Loads	EN 1998 (for all other onshore structures) EN 14620 (LNG Storage) PIANC guidelines (Port Structures)	ASCE Guidelines for Seismic Evaluation and Design of Petrochemical Facilities NFPA 59A (LNG Storage)	

NOTES: 1): If local standards, specifications, codes and regulations exist for loads, and test methods covered by this Specification that yield equivalent actions and effects, they may be substituted only with prior written approval by the Principal.

2): Note that in Canada and the US, the States/ Provincial/Territorial governments building codes, acts, and regulations can govern the national codes listed here.

3): If similar requirements exist in these codes, the most onerous loadings apply.

2. The return period of wind, wave and current loads defined in Table 3.2 shall be used in accordance with the appropriate load combination

Table 3.2 Return Periods for Wind, Wave and Current

	Return period (years)
Normal operation	1
Erection / maintenance/Test	1
Long duration construction activities (> 2 years)	10
Extreme environmental	50

3.2 STRUCTURE DEAD LOADS

1. The structure dead load shall consider all permanently attached appurtenances except the following:

   a. empty weight of process equipment, vessels, tanks, piping, and cable trays;

   b. soil and water pressure.

   Structure dead load is the weight of materials forming the structure and foundation. It includes permanently attached appurtenances to structural members e.g., lighting, instrumentation, HVAC, sprinkler and deluge systems, fireproofing, and insulation, and any additional appurtenances specified by the project.

2. For submerged structures the dry and submerged dead weight shall be calculated including the effect of the marine growth, unless the buoyancy of the marine growth is higher than the weight of the marine growth (in which case, ignore the marine growth in the dead weight calculation).

3.3 EQUIPMENT AND VESSEL DEAD LOADS

1. The empty dead load for process equipment and vessels shall be derived from the Manufacturer's/Supplier data.

   Dead load for process (rotating and static) equipment, including piping, tanks, cable trays, and vessels is the empty weight of the equipment or vessels, including all their attachments, auxiliaries, trays, internals, insulation, protective layers, fireproofing, agitators, piping, ladders, platforms.

2. Where approved vendor drawings are used as the source of information, this should be clearly indicated in the relevant structural calculations and drawings.

3.4 LIVE LOADS

3.4.1 Minimum live loads

1. Minimum live loading and reductions in live loads shall be specified as per the following order of preference (most preferred first):

   a. derived from the Manufacturer's/Supplier's data;

   b. as per the applicable codes, standards, practices and regulations listed in Table 3.1;

   c. otherwise specified by the Principal.

3.4.2 Maintenance loads

1. The minimum live loads derived in (3.4.1) shall be reviewed for their adequacy (and increased if required) in areas that are subjected to loading from access or hoisting equipment, scaffolding, and material laydown l during periodic and incidental maintenance operations.
2. Areas of flat roofs that could support mechanical equipment (e.g., HVAC) shall be designed to include loads that are produced during maintenance (i.e., by workers, equipment, materials.
3. Heat exchanger support structures and foundations that could be subject to bundle pulling (for both mechanical bundle puller and manually bundle puller) shall be designed for the horizontal load along the longitudinal axis of the heat exchanger applied at the centre of the tube bundle with a bundle pull force taken as the larger of the following:
   a. times the tube bundle weight;
   b. 9 kN (2,000 lbs) or the total weight of the exchanger, whichever is smaller.
4. For stacked exchangers, the load shall be applied at the centre of the top bundle.
5. The portion of the bundle pull load at the sliding end support shall equal the friction force or half the total bundle pull load, whichever is less:
   a. the remainder of the bundle pull load shall be resisted at the anchor end support.

3.5 WIND LOADS

1. The design shall allow for winding loading using the applicable codes.
2. Design conditions for Wind speed/load during construction and erection shall be determined by the Contractor based on the actual construction schedule and methods and subject to approval by the Principal.
3. There shall be no reductions in velocity pressure due to apparent shielding afforded by adjacent buildings and other neighbouring structures or terrain features.

3.6 SNOW LOAD, ICE LOAD, SAND LOAD AND WATER LOAD

1. The design shall allow for snow, ice, rain and sand loads using the applicable codes.
2. Maximum rainwater accumulation load shall be assessed as an accidental load.

   Rainwater accumulation occurs if the drains, drain slots, drain pipes or down spouts become blocked, or due to the roof deflection or freeze-thaw and wet snow conditions when it is likely that drains will be blocked with dense snow or ice as an incidental load.
3. On flat roofs the maximum accumulation height shall be the height of the parapet.

3.7 WAVE, CURRENT, MOORING AND BERTHING LOADS

Refer to DEP 35.00.10.10-Gen for wave, current, mooring and berthing loads on marine structures.

3.8 EARTHQUAKE LOADS

1. Refer to DEP 34.00.01.10-Gen for determining Earthquake loads.
2. Earthquake conditions/loads during construction/erection shall be determined by the Contractor and issued to the Principal for approval.

3.9 THERMAL LOADS

1. The effect of thermal loads shall be included in the design calculations.

   Sources of thermal loads are as follows:

   - *forces on vertical vessels, horizontal vessels, or heat exchangers caused by the thermal expansion of the pipe attached to the vessel;*
   - *self-straining thermal forces caused by the restrained expansion of horizontal vessels, heat exchangers, and structural members in pipe racks or in structures;*
   - *self-straining pipe anchor and guide loads;*
   - *pipe rack friction forces caused by the restrained sliding of pipes or friction forces caused by the sliding of horizontal vessels or heat exchangers on their supports, in response to thermal expansion;*
   - *restrained deflections (expansion, contraction and bending) of structural members due to annual and daily temperature variations.*

2. If thermal expansion or contraction results in friction between equipment and supports, the friction force shall be taken as the operating load on the support times the minimum friction coefficient given in Table 3.3:

   a. alternate means of reducing thermal loads due to friction shall be subject to principal's approval;

   b. the maximum sliding bearing pressures of materials referenced in Table 3.3 shall be taken into account.

3. Thermal loads and displacements shall be calculated on the basis of maximising the difference between ambient and equipment operating or installation temperature.

   a. To account for the increase in temperatures of steel exposed to sunlight, a minimum of 20 °C (35 °F) shall be added to the maximum ambient temperature.

4. In the design of pipe supporting structure, the horizontal slip forces exerted by expanding or contracting pipes on steel pipe racks shall be a minimum of 15% of the operating weight on the beam.

   a. These 'slip forces' shall not be distributed to the foundations.

5. The design of the pipe support structure, including pipe rack structure, shall include the effect of pipe friction forces and (reaction) pipe anchor forces.

6. If bellows/compensators are fitted to accommodate the expansion resulting from contraction of cryogenic piping carrying cryogenic products, the design pipe support loads should be based on the specific design codes for the bellows/compensators.

7. The horizontal friction forces on a pipe support structure shall be based on the maximum of the following, unless project specific pipe stress analysis calculations indicate a higher force:

   a. 10% of total operating weight of all lines acting on a support;

   b. 40% of total operating weight of a single largest line;

   c. 40% of total operating weight of all lines that are expanding or contracting simultaneously.

8. The friction force in (3.9, Item 7) shall be used for the design of the complete pipe support structure including its foundations.

9. Loads from anchor supports shall be combined with wind loading for the design of pipe support structure.

   Frictional forces need not be included in wind and seismic load combinations

10. A concrete pipe rack beam shall be designed for a horizontal pipe anchor force of 15 kN acting at the mid span, which is not distributed to the foundations.

11. For pipe anchor forces transferred by longitudinal girders to structural anchors (bracing), a force of 5% of the total pipe operating load per layer shall be taken into account in the design unless the pipe stress calculations dictate a higher force.

    a. These forces shall be distributed to the foundations.

Table 3.3 Minimum friction coefficients

Surfaces	Friction coefficient
Steel to Steel (not corroded)	0.4
Steel to Concrete	0.6
Proprietary sliding surfaces or coatings (e.g., PTFE)	According to Manufactures' instructions

12. Foundations and liquid retaining structures that are subject to thermo-mechanical effects shall be designed for the thermal loads.

    a. An engineering review should be carried out to assess potential changes in properties if the temperature of the concrete exceeds 70 °C (158 °F).

       Heat transfer calculations are used to determine the effects of:

       - *Thermo-mechanical forces and stresses;*
       - *Changing of any properties of materials used.*

3.10 DYNAMIC LOADS FROM ROTATING EQUIPMENT

1. Dynamic analysis shall be performed in accordance with (3.10.1) through (3.10.5) for the expected dynamic loads to design the support structures or foundations for the following:

   a. centrifugal machinery greater than 375 kW (500 HP);

   b. reciprocating machinery greater than 150 kW (200 HP);

   c. other machinery specified by the Principal.

2. The dynamics loads should be derived from information specified by the Manufacturers/Suppliers.

   In the absence of data from the Manufacturer, consult the Principal.

3. The dynamic soil properties used for final design using dynamic analysis shall be obtained from down hole or cross hole tests performed during site specific soil investigation.

3.10.1 Dynamic analysis

1. A three-dimensional dynamic analysis shall be made for rotating equipment foundations.

3.10.2 Exciting force

1. For the vibration analysis of structures and foundations of rotating equipment), the exciting forces shall be taken as the maximum values that, according to the Manufacturer/Supplier of the equipment, will occur during the lifetime of the equipment.

3.10.3 Schematic numerical model

1. The vibration calculation shall be based on a numerical model wherein the weights and elasticity of both structure and foundation and the weight of the equipment are represented.

3.10.4 Frequencies

1. Structures and foundations that support dynamic equipment shall have a natural frequency that is outside the range of 0.8 – 1.2 times the operating frequency of the equipment.
2. All natural frequencies below 2 times the operating frequency for reciprocating equipment and below 1.5 times the operating frequency for centrifugal equipment shall be calculated.
3. The damping value used shall be submitted for Principal’s approval.
4. The maximum allowable eccentricity between the centre of gravity of the combined weight of the foundation and machinery and the bearing surface shall be 5% of the respective foundation dimension in each of the length and breadth direction.

3.10.5 Dynamic amplitudes

1. Dynamic amplitudes (peak to peak amplitude) of any part of the foundation shall not exceed the lower of the following values:
   a. the maximum allowable values stated by the Manufacturer/Supplier of the equipment;
   b. 50 µm value;
   c. the calculated amplitude (single amplitude) that does not cause the effective velocity of vibration to exceed any of the following;
      i. 2 mm/s (0.08 in/s) at the location of the machine-bearing housings;
      ii. 2.5 mm/s (0.1 in/s) at any location of the structure.

      The effective velocity is defined as the square root of the average of the square of the velocity, velocity being a function of time. In the case of a pure sinusoidal function the effective velocity is 0.71 times the peak value of the velocity.
2. Any proposal to exceed a dynamic amplitude value of 50 µm shall include written acceptance by the equipment Manufacturer/Supplier.

3.11 CRANE AND HOIST LOADS

1. Design for lateral and longitudinal horizontal forces shall meet all of the following:
   a. as per crane Manufacturer/Supplier requirements;
   b. applicable code and standard listed in Table 3.1;
   c. as minimum loads in Table 3.4.
2. Crane loads shall include the maximum lifting capacity (operational capacity and test load level) as well as the maximum horizontal loads caused by wind and lateral movement (braking or acceleration).

   These specified crane loads also apply to maintenance and construction cranes (3.15) and to other lifting facilities such as monorails and hoists.
3. For the design of each structural element supporting the crane or other moving loads the load envelope shall be established to cover all positions of the crane/moving loads to maximize the design forces:
   a. applicable loads on additional facilities (e.g., crane stowage system used to secure the crane in storm wind conditions) shall be included in the design.
4. Lifting lugs or pad eyes and internal members (included both end connections) framing into the joint where the lifting lug or pad eye is located shall be designed for 100% impact load:
   a. all other structural members transmitting lifting forces shall be designed for 15% impact load;
   b. impact loads for members that support travelling cranes and hoists shall be in accordance with provisions of applicable standard(s) or code(s) listed in Table 3.1.
5. Impact loads for davits shall be the same as those for electric monorail cranes in accordance with provisions of applicable standard(s) or code(s) listed in Table 3.1.
6. Design loading for paved areas that are subject to impact from moving equipment shall be in accordance with provisions of applicable standard(s) or code(s) listed in Table 3.1.
7. Design loading for structures that are subject to impact from moving railway equipment that operates on tracks shall be increased in accordance with the provisions of the applicable standard(s) or code(s) listed in Table 3.1.
8. Allowable stresses shall not be increased when combining impact with dead load.
9. Impact loads shall be considered a variable load (similar to live load) in the selection of load factors, safety factors, and load reduction factors in the applicable load combinations.

10. Engineered lifting components shall be designed as follows:
    a. take the design lift load as two times the lifted load unless a larger factor is required by applicable codes and standards in Table 3.1. This factor includes impact;
    b. allowable stresses are according to the specification of applicable codes and standards in Table 3.1 for all loading conditions with no increase in allowable stresses such as those for temporary and environmental conditions as shown in Table 3.5;
    c. all potential orientations, load paths, and combined stress conditions that will occur throughout the lift are considered, including cable directions out of the principal plane of the components such as skewed cables;
    d. lateral thrust that can result from side-to-side sway and wind during lifting are considered;
        i. as a minimum, horizontal forces are taken as 5% of the vertical design load.
    e. lifting lugs are designed for bending about the weak axis using a force equal to a minimum of 5% of the force in the sling;
    f. the design of the attachment welds for the lifting lug consider local effects such as stress concentrations and lamellar tearing.
11. Loads imposed by the crane shall not exceed the allowable bearing capacity of the soil, per the soil report.
12. Foundations design for cranes shall be submitted for approval by the Principal.
13. Design loads applied due to cranes shall not be less than those shown in Table 3.4.

Table 3.4 Design loads Applied Due to Cranes

	Electric Operation	**Hand Operation**
Vertical impact loads - increase maximum wheel loads by:	25%	10%
Horizontal forces on rails – (taken as a percentage of the rated capacity of the crane combined with the weight of the hoist and trolley) Transverse to each rail:	10%	5%
Horizontal forces on rails – (taken as a percentage of the maximum wheel loads of the crane:) Along the rails:	10%	5%

3.12 BLAST AND IMPACT LOADS

1. Blast loads and impact loads shall be included in the design if required by local regulations, standards and codes specified in Table 3.1 or specified by the Principal.
2. Refer to DEP 34.17.10.32-Gen. for blast loading requirements.

3.13 FLUID LOADS

1. Fluid load shall be based on the maximum level and density that could occur during operation and maintenance.

   The operating contents of piping, vessels, and storage tanks including liquid, catalyst, inert balls, packing, etc., are treated as fluid loads per applicable standards or codes in Table 3.1

3.14 FLUID SURGE

1. Forces due to surging action of liquids or fluidized solids in process equipment or piping shall be considered in the design of structures.
   a. Calculations and designs to resist such forces shall be submitted to the Principal for review.

3.15 ERECTION AND TRANSPORTATION LOADS

1. Loading and supporting conditions during erection and transportation for the transported structural system and the roadways or other handling means used shall be in line with proposed erection and transportation procedures.
2. Constructability studies shall be done to determine construction loads.
3. Beams and floor slabs in multi-storey structures (e.g., fire decks) shall be designed to carry the full construction loads imposed by the props supporting the structure immediately above.
4. A note should be added to the relevant construction drawings to inform the field engineer of the adopted design/construction/erection philosophy.
5. Transportation and erection loads shall be considered as a variable load per applicable code in Table 3.1 for the selection of load factors or safety factors in the applicable load combination.
6. Transportation loads for sea going transport shall be based on the most extreme conditions from a minimum 10-year seasonal storm for the worst part of the route.
   a. Roll, Pitch, and Heave motion response of the barge/ship should be developed in conjunction with the transportation Contractor.
7. Constructability studies shall be provided for the lift load (maximum weight of equipment to be lifted), determined as follows:
   a. weight determined from manufacturing data or engineering calculations for new equipment to be lifted and increased by 10% to account for manufacturing tolerances;
   b. calculated weight for the removal of existing equipment to include the weight of all additions since installation and a minimum 25% increase factor to account for residual process materials and lifting hindrances (e.g., adhesion);
   c. weight obtained from a certified weighing, for which no increase is necessary.

3.16 TEST LOADS

1. Designs shall include loads and load combinations that occur during testing of piping, equipment and structures.
2. Test dead load shall be incorporated in the design of the supporting structure.

   Test dead load for process equipment and vessels is the empty dead load plus the weight of test medium contained in a set of simultaneously tested piping systems.
3. The test medium shall be as specified in the contract documents or as specified by the Principal.
   a. Unless the cleaning load is governing (where the cleaning fluid is heavier than the test medium), a specific gravity of 1.0 shall be used for the test medium.
4. Equipment and that segment of pipes that will be simultaneously tested shall be included for load calculations.

   If more than one piece of equipment is supported by one structure, the structure needs only to be designed on the basis that one piece of equipment will be tested at any one time, and that the others will either be empty or still in operation unless otherwise specified by the Principal.

3.17 BUOYANCY AND HYDROSTATIC PRESSURES

1. Where the bottom of a structure or equipment extends below water level, either temporary or long-term, buoyancy and hydrostatic pressures shall be accounted for in the design.
2. When evaluating the impact of buoyancy, a structure or vessel shall be considered as empty (e.g., without operating contents but including insulations).
3. The highest ground water level specified by the project shall be used for design when checking stability under earthquake loads.
4. When checking stability under wind loads, ground water level shall be taken at grade.

3.18 DIFFERENTIAL SETTLEMENT

1. The deflections, bending moments, shear and axial forces due to differential settlement shall be included in the structural design based on the selection of the foundation and as specified in soil investigation report.

   The variability of the soil strata and loading might result in differential settlement.

3.19 EARTH PRESSURE

1. Earth pressure shall be calculated for all design conditions and in accordance with the relevant geotechnical report (4).
2. The geotechnical report should include soil properties such as bulk density, cohesion and active and passive pressure coefficients.

3.20 LOAD COMBINATIONS

3.20.1 General

1. The load combinations and partial safety load factors of buildings, structures, equipment, vessels, tanks, and foundations shall be in accordance with the applicable design codes and standards for loads as specified in Table 3.1 and any other governing combination specific to the project.
2. When excluding loads other than structural, equipment and vessel dead loads results in a more critical loading condition, then such exclusion or removal of equipment and vessels shall be considered while arriving at the load combination.
3. Load factors and combinations shall be subject to the Principal's review and approval.
4. Table 3.5 provides indicate load combinations. Contractor shall propose load combinations specific to the project requirements and include it in the design basis for review and approval by Principal.

Table 3.5 Load combinations

	Load combination				
Loads	**Operation**		**Test**	**Erection**	**Empty (Shutdown)**
	Normal	**Extreme**			
	A	B	C	D	E
Dead Loads					
Dead Loads of Structure (3.2) excluding fireproofing				X	
Dead Loads of Structure including fireproofing (3.2)	X	X	X		X
Dead Loads of Empty Equipment, and Vessels excluding internals, fireproofing, and insulation (3.3)				X 5)	
Dead Loads of Empty Equipment, and Vessels including Internals, Fireproofing, and Insulation (3.3)	X	X	X		X
Test and Maintenance Loads					
Weight of Test Medium (3.16)			X		
Maintenance Loads (3.4.2)					X 8)
Crane and Hoist Loads (3.11)			X	X	X
Live Loads					
Platform and Walkway Loads (3.4.1)	X	X	X		X
Material Storage Loads	X	X	X		
Crane and Hoist Loads (3.11)	X	X			

	Load combination				
Loads	**Operation**		**Test**	**Erection**	**Empty (Shutdown)**
	Normal	**Extreme**			
Environmental Loads					
Wind Loads (3.5)	X	X	X[6),4)]	X[4)]	X[7)]
Snow/Sand/Water Loads (3.6)	X	X		X	X
Wave and Current Loads (3.7)	X[6)]	X[6)]	X[6)]	X[6)]	X
Earth Pressure (3.19)	X	X	X	X	X
Earthquake Loads (3.8)	X	X			X[7)]
Process Loads					
Dynamic Loads (3.10)	X	X	X[2)]		
Blast and Impact Loads (3.12)	X	X			
Thermal Loads (3.9)	X	X			
Normal Fluid Loads (3.13)	X				X[9)]
Start-up/Shutdown Fluid Loads (3.13)		x			
Normal Surge Loads (3.14)	X[1)]				
Extreme Surge Loads (3.14)		X[1)]			
Erection and Transportation Loads					
Loads during Erection (3.15)				X	
Other Loads					
Load due to Differential Settlement (3.18)	X	X	X		X
Buoyancy and Hydrostatic Pressures (3.17)	X	X			X

NOTES:

1) Surge loads are treated as variable loads.
2) Only if the structure supports rotating equipment that will be in operation while a vessel is being tested with water.
3) Deleted
4) The effect of wind and earthquake forces acting on temporary scaffolding erected during construction or for subsequent maintenance which will be transferred to the vessel or column is considered.
   When considering wind effects, the actual projected area of the scaffold members together with the correct shape factor and drag coefficient should be used. As an initial approximation, the overall width of the scaffolding itself can be taken as 1.5 m on each side of the vessel or column with 50% closed surface and shape factor 1.
5) Take account of the empty weight of equipment during erection.
6) Refer (3.1)
7) Wind and earthquake are not considered to act simultaneously.
8) Bundle pull load is not combined with wind or earthquake.
9) Used for cleaning media.

3.20.2 Allowable or working stress design

1. Increases in allowable stresses for designs using Allowable Stress Design shall not be used with the loads or load combinations unless it can be demonstrated to the Principal that such an increase is justified by structural behaviour caused by rate or duration of load.

3.20.3 Limiting state or strength design

1. Partial Load safety factors as per codes specified in Table 3.1 shall be used in establishing all critical load combinations.

4. DATA GATHERING, SITE INVESTIGATION AND FOUNDATION ENGINEERING

Data gathering is required to obtain sufficiently reliable information on the local conditions for seismic loads and environmental conditions (such as wind, waves and currents).

Local standards might provide such information, but a site specific meteorological data gathering campaign might be required to collect data.

1. A site specific geotechnical investigation shall be carried out to determine the character and variability of the soil strata underlying the foundations of the proposed structures.
2. Refer to DEP 34.11.00.10-Gen. for guidelines for the determination of the extent and scope of site investigations and associated reports.

5. STRUCTURE DESIGN CRITERIA

5.1 GENERAL

1. Refer to DEP34.28.00.31-Gen. for structural design criteria for steel structures.
2. Refer to DEP 34.19.20.31-Gen. for structural design criteria for concrete structures.
3. For Composite Steel & Concrete and Masonry the applicable standards and codes listed as per Table 5.1 below shall be followed.

Table 5.1 Applicable Codes, Standards and Practices for Structure Design Criteria

Material/ Structure Type	European Codes and Standards	American Codes and Standards	Canadian Codes and Standards
Composite Steel and Concrete	EN 1994	ACI 318/ 318R and ANSI / AISC 360	CSA S16 and CSA 23.3
Masonry	EN 1996	ACI 530 ASCE 5 TMS 402	CSA S304.1

5.2 FOUNDATIONS

1. Foundation designs shall be based on the results of a geotechnical engineering investigation as specified in (4).
2. The soil bearing capacity and pile capacity in compression, lateral and tension shall be as recommended by the geotechnical consultant, subject to the Principal approval.
3. The foundation design shall address the effects of long-term and differential settlement, if applicable.
4. Refer to (5.4) for requirements for foundation for rotating and reciprocating equipment.
5. Foundations for static (non-rotating or non-reciprocating) equipment, buildings and structures shall meet the requirements of one of the following:
    a. EN 1997 and EN 1998;
    b. ASCE/SEI 7-05 supplemented by the ASCE Guidelines for Seismic Evaluation and Design of Petrochemical Facilities;
    c. National Building Code supplemented by the ASCE Guidelines for Seismic Evaluation and Design of Petrochemical Facilities.
6. The Contractor shall ensure that equipment supports, buildings and structures can accommodate the resultant long-term and differential settlement from foundations.
7. The foundations shall be designed to meet the settlement limitations of the supported assets.
8. The top of grout (bottom of base plate) of pedestals and ringwalls shall be 300 mm (1 ft) above the high point of finished grade.
9. Refer DEP 34.51.01.33-Gen. or DEP 34.51.01.31-Gen for Tank foundations and their anchorage.
10. Refer DEP 34.11.00.11-Gen. for site preparation and earth works.

    Except for foundations supporting ground-supported storage tanks, uplift load combinations containing earthquake loads need not include the vertical components of the seismic load effect, if seismic is used to size the foundation.

5.3 MAXIMUM ALLOWABLE HORIZONTAL DRIFT AND VERTICAL DEFLECTION

5.3.1 Maximum allowable horizontal wind drift

1. The maximum horizontal wind drift shall not exceed the following limits given in Table 5.2.

Table 5.2 Maximum allowable horizontal wind drift

Case #	Case	Maximum Allowable Horizontal Wind Drift[1]
1	Wind drift for pipe racks	H/100
2	Wind story drift for occupied buildings	H/200
3	Wind drift for pre-engineered metal buildings	H/80
4	Wind drift for buildings with a bridge crane that is required to be in services during cyclones/hurricanes conditions	H/400 or 50 mm (2 in)
5	Wind drift for buildings with a bridge crane that is not required to be in service during cyclones/hurricanes conditions	H/140 or 50 mm (2 in)
6	Wind drift for process structures and personnel access platform	H/200 [2)]

NOTES: 1) H = the structure (e.g., pipe racks or buildings) height applicable for case 1 to case 5
2) H = the structure height at elevation of drift consideration

5.3.2 Maximum allowable vertical deflection for crane supports

1. Non-impact vertical deflection of support runway girders shall not exceed the limits given in Table 5.3 if they are loaded with the maximum wheel load(s) or the limits specified by the crane Manufacturer/Supplier.

Table 5.3 Maximum allowable vertical deflection for girders

Case #	Case [2)]	Maximum Allowable Vertical Deflection [1)]
1	Top-running infrequent service, light service or moderate service (e.g., CMAA class A, B, and C cranes or equivalent)	L/600
2	Top-running heavy service (e.g., CMAA class D cranes or equivalent)	L/800
3	Top-running severe service or continuous severe service (e.g., CMAA class E and F cranes or equivalent)	L/1000
4	Under-running infrequent service, light service or moderate service (e.g., CMAA class A, B, and C cranes or equivalent)	L/450
5	Monorails	L/450

NOTES: 1) L = the span length
2) CMAA stands for Crane Manufacturers Association of America.

2. The vertical deflection of jib crane support beams shall not exceed L/225 if loaded with the maximum lifted plus hoist load(s), without impact.

   where L = the maximum distance from the support column to load location along the length of the jib beam.

3. The lateral deflection of support runway girders for cranes with lateral moving trolleys shall not exceed L/400 (where L = the span length) if loaded with a total crane lateral force not less than 20% of the sum of the weights of the lifted load (without impact) and the crane trolley.

   a. The lateral force shall be distributed to each runway girder with consideration for the lateral stiffness of the runway girders and the structure supporting the runway girders.

4. Crane stops shall be designed in accordance with the crane Manufacturer/Supplier's requirements:

   a. the design force according to Equation 5.1 should be considered in the absence of vendor's supplied data.

$$F = W\,V^2 / (2gTn) \qquad \text{Equation 5.1}$$

where:

F = Design force on crane stop, kN (kips).

W = 50% of bridge weight + 90% of trolley weight, excluding the lifted load, kN (kips).

V = Rated crane speed, m/sec (ft/sec).

g = Acceleration of gravity, 9.8 m/sec^2 (32.2 ft/sec^2).

T = Length of travel of spring or plunger required to stop crane, from crane Manufacturer/Supplier, typically 0.05 m (1.15 ft).

n = Bumper efficiency factor (0.5 for helical springs of as advised by the crane Manufacturer/Supplier for hydraulic plungers).

5.4 SUPPORTS FOR ROTATING EQUIPMENT

1. For centrifugal machinery less than 500 horsepower, in the absence of a detailed dynamic analysis, the foundation weight shall be designed to be at least three times the total machinery weight, unless specified otherwise by the equipment Manufacturer/Supplier.

   For vertical pumps the mass of the motor may be assigned to the mass of the foundation.

2. For reciprocating machinery less than 200 horsepower, in the absence of a detailed dynamic analysis, the foundation weight shall be designed to be at least five times the total machinery weight, unless specified otherwise by the Manufacturer/Supplier.

3. The soil-bearing or pile capacity for foundations for equipment designed for dynamic loads shall be as specified by DEP 34.11.00.12-Gen.

4. Foundations for rotating, reciprocating equipment or foundation for structures that support these equipment shall be designed in accordance with API RP 686 Chapter 4.

5.5 GUY WIRES

1. Guy wires shall have a safety factor of at least 5 against failure due to breakage or pull-out of anchorage.

5.6 EXISTING STRUCTURES

1. If the Principal agrees that the integrity of the existing structure is at least 100% of the original capacity without new loads, based on the design code in effect at the time of original design, structural designs shall be performed in accordance with the following:
   a. if additions or alterations to an existing structure do not increase the force in any structural element or connection by more than 5%, no further analysis is required;
   b. original design and drawings shall be updated with changes due to new equipment/loads;
   c. the strength of any structural element or connection is not decreased to less than that required by the applicable design code or standard for new construction for the structure in question;
   d. if the increased forces on the element or connection are greater than 5%, the element or connection is analysed to show that it is in compliance with the applicable design code for new construction, considering the following;
      i. available data including assumptions used for the loads, load combinations, codes and standard previously used in the original design;
      ii. assessment of material strength of the existing structure, i.e., steel, concrete, anchor bolts, connection bolts, welding.
   e. Update As-built drawings to reflect any strengthening to the existing structure and new design envelopes under which the structure is expected to operate.

6. REFERENCES

In this DEP, reference is made to the following publications:

NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
2. The DEPs and most referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at http://sww.shell.com/standards/.

SHELL STANDARDS

DEP feedback form	DEP 00.00.05.80-Gen.
The use of SI quantities and units (endorsement of ISO/IEC 80000)	DEP 00.00.20.10-Gen.
Preparation of technical drawings	DEP 02.00.00.10-Gen.
Earthquake design for onshore facilities – Seismic hazard assessment	DEP 34.00.01.10-Gen.
Site investigations	DEP 34.11.00.10-Gen.
Site preparation and earthworks including tank foundations and tank farms	DEP 34.11.00.11-Gen.
Design of blast resistant onshore buildings, control rooms and field auxiliary rooms	DEP 34.17.10.30-Gen.
Vertical steel storage tanks - Selection design and construction (amendments/supplements to EN 14015)	DEP 34.51.01.31-Gen.
Aboveground vertical storage tanks (amendments/supplements to API Standard 650)	DEP 34.51.01.33-Gen.
Design of jetty facilities (amendments/supplements to BS 6349-1, 1.1/1.3/1.4, BS 6349-2 and BS 6349-4)	DEP 35.00.10.10-Gen.

AMERICAN STANDARDS

American Association of State Highway and Transportation Officials - Standard Specifications for Highway Bridges (HB-17)	AASHTO
American Concrete Institute - Building Code Requirements for Structural Concrete (ACI 318-11) and Commentary	ACI 318/318R
American Concrete Institute - Building Code Requirements and Specification for Masonry Structures	ACI 530/ASCE 5/TMS 402
Specification for Structural Steel Buildings	ANSI/AISC 360
American Petroleum Institute - Recommended Practice for Machinery Installation and Installation Design, 2nd edition 2009	API RP 686
American Society of Civil Engineers Guidelines - Seismic Evaluation and Design of Petrochemical Facilities	ASCE Guidelines
American Society of Civil Engineers Guidelines - Wind Loads and Anchor Bolt Design for Petrochemical Facilities	ASCE Guidelines
American Society of Civil Engineers - Minimum Design Loads for Buildings and Other Structures	ASCE/SEI 7-05
American Society of Civil Engineers - Design Loads on Structures During Construction	ASCE/SEI 37-02
Specifications for Top Running Bridge and Gantry Type Multiple Girder Overhead Electric Travelling Cranes	CMAA No. 70

Specifications for Top Running and Under Running Single Girder Electric Overhead Traveling Cranes Utilizing Under Running Trolley Hoist	CMAA No. 74
Issued by: Crane Manufacturers Association of America, Inc.	
Standard for the production, storage, and handling of liquefied natural gas (LNG)	NFPA 59A
US Occupational Safety & Health Administration (OSHA) – Part 1910: Occupational Safety and Health Standard 179: Overhead and gantry cranes	OSHA 1910.179

CANDIAN STANDARDS

Design of steel structures	CSA S16
Design of concrete structures	CSA 23.3
Design of masonry structures	CSA S304.1
National Building Code of Canada	NBC

BRITISH STANDARDS

Maritime works – Part 2: Code of practice for the design of quay walls, jetties and dolphins	BS 6349-2

EUROPEAN STANDARDS

Euro code 1 – Actions on structures	EN 1991
Part 1-6: General actions - Actions during execution	EN 1991-1.6
Part 1-7: General actions - Accidental actions	EN 1991-1.7
Euro code 3 – Design of steel structures	EN 1993
Part 6: Crane supporting structures	EN 1993-6
Euro code 4 – Design of composite steel and concrete structures	EN 1994
Euro code 6 – Design of masonry structures	EN 1996
Euro code 7 – Geotechnical design	EN 1997
Euro code 8 – Design of structures for earthquake resistance	EN 1998
Design and manufacture of site built, vertical, cylindrical, flat-bottomed steel tanks for the storage of refrigerated, liquefied gases with operating temperatures between 0 °C and –165 °C	EN 14620

INTERNATIONAL STANDARDS

International Code Council (ICC): International Building Code®	IBC
PIANC MarCom Report of WG 34 – Seismic design guidelines for port structures	PIANC
Petroleum and Natural Gas Industries – Specific Requirements for offshore structures	ISO 19901
Petroleum and Natural Gas Industries – Fixed Steel Offshore Structures	ISO 19902

APPENDIX A DOCUMENTATION REQUIRED DURING DESIGN STAGES

A.1 DESIGN STAGES

1. The units of measure in all calculation and drawing deliverables shall be subject to approval by the Principal.
2. All the computer programs to be used by the Contractor shall be subject to prior approval of the Principal.
   a. All required documentation shall be supplied to demonstrate their accuracy and applicability.
3. When computerised software is used to perform the basic design the Contractor shall submit electronic copies of files of the following:
   a. Input data;
      i. graphical or numerical isometrics, plans and sections print outs of showing both joints numbers and members numbers;
      ii. graphical or numerical isometric, plans and section print outs showing member sizes;
      iii. graphical or numerical isometrics for each load case;
      iv. loads;
      v. loading combinations;
      vi. connection types;
      vii. unbraced lengths of members.
   b. Output data;
      i. basic load reactions;
      ii. envelope reactions for all foundation loading combinations only;
      iii. envelope stress ratio for all steel members;
      iv. end connections forces reactions;
      v. drift and deflection values;
      vi. sum of reaction forces for each load case.

A.1.1 Basic design

A.1.1.1 General

1. Before the detailed design is started, a basic design documentation shall be submitted consisting of the following:
   a. a basic sketch (A.1.1.2);
   b. design statement and calculation (A.1.1.3);
   c. design of main structural components (A.1.1.4).

A.1.1.2 Basic sketch

1. The sketch shall show the proposed structure (in an isometric perspective, plan, elevation and a series of cross sections) including foundations and/or anchorage:
   a. isometric perspectives may be added for clarity;
   b. structural members may be shown as single lines.
2. Main dimensions and materials of construction should be defined on the sketch.
3. Loads in the main structural members shall be classified in accordance with (3) and to be shown on a sketch.

A.1.1.3 Design statement and calculation

1. The design statement shall document the design philosophy/choices and summarise all relevant starting points for the calculations, such as load data, supporting conditions, design criteria, code references, applicable theories, methods of analyses, assumptions, clarifications.
2. The calculations shall take into account the soil investigation report (4).

A.1.1.4 Main structural members

1. Typical details that determine the feasibility of the concept shall be shown, such as the connections between the various elements in case prefabricated concrete elements are used the connections.

   Standard structural details, such as connections of steel beams and columns or details of reinforcing steel over the full length of a reinforced concrete beam need not be shown
2. All structural details, such as connections between the various elements in case prefabricated concrete elements are used shall be shown.
3. When prefabricated concrete elements are used the connections between the various elements shall be designed and calculations submitted for approval.
4. The calculations shall state the loads in the main structural members (axial loads, bending moments, shear and possibly torsion), and include the reaction forces on the foundation/anchorage (load per pile or per unit of area).

A.1.2 Detailed design documentation

1. The detailed design documentation with the following table of contents shall be submitted to the Principal for approval:
    a. a cover page with a report title, number, date and revision;
    b. summary and conclusion;
    c. the table of contents;
    d. the lists of Figures and Tables;
    e. introduction;
    f. design philosophy;
    g. applicable codes, formulas, graphs/tables;
    h. references to literature, etc., for subjects not covered by applicable codes;
    i. loading tables with loading location diagrams;
    j. if computer programs are used, the following information are supplied;
        i. logic and theory used;
        ii. analytical model of the structure used for computer analysis;
        iii. a manual calculation to prove the validity of the computer analysis;
        iv. input data such as geometry, supports and calculated loads and load combinations;
        v. output data such as member forces and deflections.
    k. The material, quality and section selection of the members;

APPENDIX B DRAWINGS AND RELATED DOCUMENTS

1. Drawings shall be electronically prepared and stored.
2. Refer DEP 02.00.00.10-Gen for the standard sizes requirement.
3. Drawings shall comply with the following minimum requirements:
    a. each layout drawing to include a grid indicating Northings and Eastings;
    b. plant grid to be used for onshore layouts;
    c. true North and Plant indication and their relation is included on all layout drawings;
    d. refer DEP 00.00.20.10-Gen for Dimensions on the drawings that are in SI units;
    e. on all drawings levels are indicated in metres (feet) against Plant Datum, all other dimensions in millimetres (feet-inches);
    f. on all drawings the relation between Plant Datum and Local Land Datum is indicated;
    g. bathymetric drawings include water depth contours;
    h. depth contours are indicated in metres (feet-inches) relative to Chart Datum;
    i. the drawings include a reference between Chart Datum, Local Land Datum and Plant Datum;
    j. the drawing numbers of all reference drawings are included on each drawing;
    k. the Unit of dimensions are indicated on each drawing;
    l. the drawing numbers and revisions are included on each drawing with the following as a minimum;
        i. principal's drawing number and CAD number;
        ii. the CAD number complying with the filename of the electronic version of the drawing.
    m. all text in English;
    n. each drawing bears the following information, in the bottom right-hand corner;
        i. order number of the Principal;
        ii. name of plant (e.g., catalytic cracking unit);
        iii. name of unit (e.g., compressor building);
        iv. name of part of the unit (e.g., portal frames);
        v. copyright clause (to be advised by the Principal).
    o. only drawings marked "Released for construction" by the Contractor responsible for design and engineering to be used at the site;
    p. drawings are submitted to the Principal for review and/or approval together with the relevant calculations, including those required for submission to local authorities.
4. Material quantity take off should be mentioned in Approved for Construction (AFC) drawings.
5. In case of revised drawings or documents, the section revised shall be clearly marked.
6. Items on hold shall be clearly identified and marked on drawings.